

Question	Marking Guidance	Mark	Comments
10.1	1. Amino acids joined by peptide bond(s); 2. (By) condensation reaction(s); 3. Secondary structure is formed by hydrogen bonding; 4. Tertiary structure formed by interactions (between R groups); 5. Quaternary structure contains >1 <u>polypeptide</u> OR Quaternary structure formed by interactions/bonds between <u>polypeptides</u> ;	5 (5 x AO1)	3. Accept alpha helix OR β -pleated sheet for 'secondary structure' 4. Accept 3° for tertiary 4. and 5. Accept for 'interactions', hydrogen bonds OR disulfide bridges OR ionic bonds OR hydrophobic OR hydrophilic interactions 5. Ignore peptide

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10.2	1. Polymer of nucleotides; 2. (Nucleotide) consists of deoxyribose, phosphate and an organic/nitrogenous base; 3. Phosphodiester bonds (between nucleotides); 4. DNA double helix held by H bonds OR 2 strands held by H bonds; 5. (Hydrogen bonds/pairing) between adenine, thymine and cytosine, guanine; 6. DNA is associated with histones/proteins; 7. (During mitosis/when visible) chromosome consists of two chromatids joined at a centromere;	6 max (6 x AO1)	1. Accept 'polynucleotide' 2. Accept 'phosphoric acid' for phosphate 5. Ignore bases identified with letters (A, T, G, C) 5. Reject adenosine and cysteine Accept correctly annotated diagram(s) for equivalent marking points

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10.3	1. Independent segregation of homologous Chromosomes/pairs; 2. Crossing over between homologous chromosomes/pairs; 3. Random fertilisation of gametes; 4. (Produces) new combinations of <u>alleles</u> ;	4 (4 x AO1)	If no marks awarded accept one principle mark 'Meiosis producing cells that are genetically different (from one another)' 1. For 'independent' accept 'random' 1. For 'segregation' accept 'assortment' 2. Accept 'within bivalent' for 'between homologous pair' 3. Ignore 'random mating' 3. Accept 'random fusion' for 'random fertilisation' Accept as an additional mark point 5. (Produces) new combinations of maternal and paternal chromosomes Ignore reference to epigenetics